Renal Transplant, Pediatric

ORG: P-1015 (ISC)
Link to Codes

MCG Health
Inpatient & Surgical Care
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Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Procedure

- Procedure is indicated for **ALL** of the following(1)(2)(3)(4)(5)(6):
  - End-stage renal disease, as indicated by **1 or more** of the following:
    - Need for chronic renal replacement therapy due to irreversible renal disease
    - Glomerular filtration rate 30 mL/min/1.73m^2 (0.50 mL/sec/1.73m^2) or less eGFR - Pediatric Calculator and irreversible renal disease
  - No contraindications present, as indicated by **ALL** of the following(1):
    - No multiple myeloma or monoclonal immunoglobulin deposition disease (light or heavy chain deposition) unless in stable remission
    - No amyloidosis with extensive extrarenal involvement (eg, cardiac amyloidosis)
    - No decompensated cirrhosis unless candidate for combined liver-kidney transplant
    - No severe irreversible obstructive or restrictive lung disease (eg, oxygen dependent irreversible disease)
    - No severe uncorrectable symptomatic heart disease (eg, New York Heart Association functional class IV heart failure, coronary artery disease, valvular disease) unless candidate for combined heart-kidney transplant or evaluated by a transplant cardiologist
    - No progressive central neurodegenerative disease (eg, Alzheimer, Parkinson, Huntington diseases)
    - No unstable psychiatric disorder that affects decision making
    - No ongoing substance use disorder that affects decision making
    - No inability to comply with ongoing medical care
    - No active bacterial, viral, parasitic, or fungal infection other than hepatitis C
    - No stroke within the last 6 months or transient ischemic attack within the last 3 months
    - No active gastrointestinal disease including symptomatic peptic ulcers, diverticulitis, inflammatory bowel disease, or gallbladder disease
    - No acute pancreatitis within the last 3 months
    - No acute hepatitis (eg, elevated INR or transaminitis)
    - No active malignancies[A]
    - No surgical contraindications (eg, urologic or vascular problems)

Alternatives to Procedure

- Alternatives include(9):

- Hemodialysis
- Peritoneal dialysis

Operative Status Criteria

Goal Length of Stay: 5 days postoperative

- Inpatient

Preoperative Care Planning

- Preoperative care planning needs may include(1)(2)(3)(4)(5)(6):
  - Preoperative evaluation, including:
    - Multidisciplinary renal transplant team evaluation for(1):
      - Alternatives to renal transplant
      - Contraindications to renal transplant
      - Academic assessment
      - Psychosocial evaluation(10)
    - Assessment of medical risk[B](1)(4)(5)
    - Protocol as specified by transplant center
    - Education of patient and dialysis unit staff
    - Routine preoperative evaluation. See Preoperative Education, Assessment, and Planning Tool SR.
  - Diagnostic test scheduling, including:
    - Laboratory testing
    - Detailed infectious disease screening, including latent TB, cytomegalovirus, hepatitis B and C, MMR, varicella, HIV with viral load and CD4 count if HIV positive, and antibody titers
    - Immunology screening, including blood and tissue typing and HLA antibodies
    - Nutrition evaluation(11)
    - Dental evaluation
    - Neurocognitive and academic assessment(1)
    - Chest x-ray
    - Renal imaging (renal ultrasound, CT scan, or MRI)(12)(13)
    - Urodynamic evaluation (eg, voiding dysfunction, neurogenic bladder, suspected noncompliant bladder)(14)
    - Brain MRI for patient with autosomal dominant polycystic kidney disease (to rule out intracranial aneurysm)(15)
  - Preoperative treatment, procedures, and stabilization, including:
    - Immunization as appropriate (eg, hepatitis, varicella-zoster, pneumococcus, influenza, COVID-19, routine childhood vaccinations)(16)(17)
    - Desensitization for live-donor recipients with high HLA sensitivity, HLA mismatch, or ABO mismatch(8)(18)(19)
    - Weight loss counseling if indicated(1)
    - Bladder augmentation or urinary diversion procedures as indicated (eg, children with obstructive uropathy)(14)
    - Pretransplant native kidney nephrectomy if indicated[C](1)(14)(20)
  - Preoperative discharge planning as appropriate. See Discharge Planning in this guideline.

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
1	• OR to ICU • Social Determinants of Health Assessment • Discharge planning	• **Clinical Indications met[D]**	• Bed rest	• IV fluids • IV medications	• Dialysis • Central lines • Urethral catheter • CBC • Electrolytes and renal function tests	• Immunosuppressive medication and corticosteroids • Multimodal analgesia, possible PCA • Perioperative antibiotics

2	• ICU • Social Determinants of Health Assessment	• **Hemodynamic stability**	• Bed rest	• IV fluids • IV medications	• CBC • Electrolytes and renal function tests	• Immunosuppressive medication and corticosteroids • Multimodal analgesia • Antibiotics absent
3	• ICU • Possible floor • Social Determinants of Health Assessment	• **Renal function adequate**	• **Out of bed**	• Possible clear liquid diet and oral medications • IV or oral fluids and medications	• CBC • Electrolytes and renal function tests • Immunosuppressive medication monitoring	• Immunosuppressive medication and corticosteroids • Multimodal analgesia
4	• Floor • Social Determinants of Health Assessment	• **No evidence of ileus or bowel obstruction**	• **Ambulatory[E]**	• **Clear liquid diet, advance as tolerated**	• **Urethral catheter absent** • Renal scan • Immunosuppressive medication monitoring • CBC • Electrolytes and renal function tests	• Immunosuppressive medication and corticosteroids • Multimodal analgesia • Begin oral corticosteroid taper
5	• Floor • Social Determinants of Health Assessment	• **Pain absent or managed**	• Ambulatory	• **Oral hydration** • **Oral medications** • **Advance diet**	• **Central lines absent** • Immunosuppressive medication monitoring • CBC • Electrolytes and renal function tests	• **PCA absent[F]** • Immunosuppressive medication and corticosteroids
6	• Social Determinants of Health Assessment • Floor to discharge[G] • Complete discharge planning	• **No evidence of graft rejection** • **No evidence of postoperative or surgical site infection** • **Renal function adequate** • **Diet tolerated** • **Pain absent or managed** • **Discharge plans and education understood**	• **Ambulatory or acceptable for next level of care[E]**	• **Oral hydration[H]** • **Oral medications or regimen acceptable for next level of care** • **Oral diet or acceptable for next level of care**	• Immunosuppressive medication monitoring • CBC • Electrolytes and renal function tests	• Oral immunosuppressive medication and corticosteroids

(3)(4)(6)(21)(22)(23)(24) N

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 5 days postoperative

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(3)(4)(21)(24):
  - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
    - Expect brief stay extension.
  - Complications of procedure(5)(25)
    - Delayed graft function may require adjustment of immunosuppression, hemodialysis, Doppler ultrasound, renogram, or biopsy.(26)(27)
    - Surgical complications may require renal imaging procedures, interventional radiology procedures, cystoscopic procedures, or reoperation; examples include(13)(28)(29):
      - Urine leaks and urinomas
      - Ureteral stricture or obstruction
      - Intra-abdominal or retroperitoneal hemorrhage or hematoma
      - Lymphocele
      - Abscess
      - Renal artery stenosis
      - Renal or iliac artery thrombosis(29)
      - Renal vein thrombosis(29)
      - Renal arterial injury (eg, dissection, pseudoaneurysm, arteriovenous or arteriocalyceal fistula)
      - Poor graft function
    - Severe medical complications, such as acute perioperative cardiac events, respiratory failure, or sepsis from a postoperative infection, may require continued ICU care.
    - Expect brief to moderate stay extension.
  - Acute rejection(30)(31)
    - Acute rejection presents as rise in creatinine with or without reduction in urine output.
    - Anticipate serologic testing for anti-HLA antibodies, urine or serum molecular biomarker assays, or biopsy of transplanted kidney.
    - Expect treatment with corticosteroids, immunosuppression, immunoadsorption, antibody infusion, IV immunoglobulin, antithymocyte globulin (rATG), or plasmapheresis.
    - Expect moderate stay extension.
  - Simultaneous native kidney nephrectomy[C](14)(20)
    - Native nephrectomy may be required for patient with renal tumor, symptomatic or infected polycystic kidney disease, refractory nephrotic syndrome, hypertension, anti-glomerular basement membrane antibodies, high-volume stone disease, recurrent pyelonephritis, or vesicoureteral reflux.
    - Expect moderate stay extension.
  - Care for comorbidities
    - Patient with complex comorbidities may require continued acute care.
    - Patient with diabetes may require major readjustments in insulin dosing due to changing renal insulin clearance.(32)
    - Expect brief stay extension.

See Common Complications and Conditions ISC for further information.

Hospital Care Planning

- Hospital evaluation and care needs may include(3)(4)(6)(21)(22):
  - Diagnostic test scheduling and completion, including:
    - Laboratory tests, including CBC, electrolytes, serum calcium, magnesium, phosphate, creatinine, and urine protein excretion
    - Renal imaging(13)(28)(33)
  - Treatment and procedure scheduling and completion, including:
    - IV antibiotics
    - Immunosuppressive medication, antibody induction therapy, and corticosteroids(34)(35)(36)
    - Post-transplant desensitization for live-donor recipients with high HLA sensitivity, HLA mismatch, or ABO mismatch (eg, plasmapheresis, immunoadsorption, IV immunoglobulin, rituximab)(18)
    - Serial postoperative CBCs and renal function tests
    - Urinary catheter care and removal
    - Post-transplant oral prophylactic agents (eg, for *Pneumocystis jirovecii*, cytomegalovirus, and urinary tract infection)
  - Consultation, assessment, and other services scheduling and completion, including:
    - Nephrology consultation
    - Urology consultation
    - Nutrition consultation(37)
  - Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool SR); key items include(6)(10):
    - Pain management, including medication by PCA. See Pain Management SR for further information.
    - Wound management, observing for healing, drainage, bleeding, and signs of infection. See Wounds Due to Surgery, Injury, or Pressure SR for further information.
    - Hemodynamic stability

- Fluid and electrolyte balance
- Careful monitoring for signs of rejection, including oliguria, anuria, fever, flank tenderness or swelling, weight gain, hypertension, decreased renal function, or malaise
- Respiratory status
- Neuropsychiatric problems, including delirium, delusions, emotional inappropriateness, and depression

Discharge

Discharge Planning

- Discharge planning includes[I]:
  - Assessment of needs and planning for care, including(39)(40):
    - Develop and modify treatment plan (involving multiple providers) as needed.
    - Evaluate and address preadmission functioning as needed.
    - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment SR for further information.
    - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool SR for further information.(38)
    - Evaluate and address patient or caregiver preferences as indicated.
    - Identify skilled services needed at next level of care, with specific attention to:
      - Genitourinary status assessment(21)
      - Medication management, adherence instruction, and side effects assessment(3)(41)(42)(43)
      - Pain management
      - Wound or dressing management(41)
  - Early identification of anticipated discharge destination; options include(40)(44):
    - Home, considerations include:
      - Home safety assessment. See Home Safety Assessment SR for further information.
      - Patient safe to go home; examples include(45)(46)(47):
        - Medical status stable for patient's condition
        - Functional care can safely be provided with available resources.
        - Mental status stable for patient's condition
        - Medication availability confirmed and reconciliation complete
        - Patient/caregiver education completed with written discharge instructions provided
        - Community resources identified and referrals made, as needed
        - Home care arranged, if indicated
        - Necessary medical equipment delivery arranged or available in home, if indicated
        - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
      - Access to follow-up care
      - Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment SR for further information.
      - Caregiver need, ability, and availability
    - Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool SR for further information.
  - Transitions of care plan complete, including(40)(44)(48):
    - Patient and caregiver education complete. See Renal Transplant, Pediatric: Patient Education for Clinicians SR for further information.
      - See Teach Back Tool SR for further information.
    - Medication reconciliation completion includes(43)(49):
      - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
      - Assess each medication for correlation to disease state or medical condition.
      - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
      - Provide reconciled medication list to all treating providers.
      - Confirm that patient or caregiver can acquire medication.
      - Educate patient and caregiver.
        - Provide complete medication list to patient and caregiver.
        - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
        - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(50)
      - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
      - See Medication Reconciliation Tool SR for further information.
    - Plan communicated to patient, caregiver, and all members of care team, including(51):

- Inpatient care and service providers
- Primary care provider
- All post-discharge care and service providers

- Appointments planned or scheduled, which may include:
  - Primary care provider
  - Specialists for management of comorbidities as needed(3)
  - Transplant center or team
  - Other
- Outpatient testing and procedure plans made, which may include:
  - Laboratory testing
  - Other
- Referrals made for assistance or support, which may include:
  - Community services
  - Educational program (eg, chronic condition management, self-management)(52)
  - Financial, for follow-up care, medication, and transportation
  - Self-help or support groups(53)
  - Tobacco use treatment
  - Other
- Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include:
  - Blood pressure cuff
  - Child safety seat(54)
  - Weight scale
  - Wound care equipment and supplies(55)
  - Other

Discharge Destination

- Post-hospital levels of admission may include:
  - Home.
  - Home healthcare. See Home Care Indications for Admission Section HC in Transplant, Pediatric guideline in Home Care.

Evidence Summary

Criteria

The evidence for the clinical indications found in this guideline includes 5 published peer reviewed articles, 1 specialty society or other evidence-based guideline, and 2 book sections.

Allografts from living donors are associated with better graft and patient survival rates.(5) **(EG 2)** Therefore, strategies such as kidney-paired donation and donor chains have been established to allow for matching between sets of incompatible patient-donor pairs in order to increase the utilization of live-donor transplants.(7)(8) **(EG 2)**

Length of Stay

Analysis of national hospital discharge data for a pediatric population shows 24% of hospitalized patients undergoing the principal procedure of renal transplant discharged in 5 days or less.(24) **(EG 3)**

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Footnotes

[A] Treated and cured cancers, including any carcinoma in situ, nonmetastatic basal cell and squamous cell carcinoma of the skin, melanoma in situ, prostate cancer with Gleason score 6 or less, small (3 cm or less) renal tumors, superficial bladder cancer, and follicular or papillary thyroid cancer less than 2 cm with low-grade histology, are not contraindications to transplant. Patients with hematologic or solid tumor malignancies in remission and patients with myelodysplasias, chronic leukemia, or low-grade lymphomas may be transplant candidates after evaluation by a transplant hematologist.(1) [A in Context Link 1]

[B] Assessment of medical risk in children may include evaluation for recurrent kidney disease, especially abnormalities in bladder emptying, Wilms tumor, hemolytic uremic syndrome, infection, cancer, membranous or membranoproliferative glomerulonephropathy, and coagulopathy or hypercoagulable state.(1)(2)(4)(5) [B in Context Link 1]

[C] Indications for native nephrectomy may include polycystic kidney disease with severe recurrent complications (eg, bleeding, infection, stones, intractable pain), severe hypertension resistant to medical management, severe proteinuria causing coagulopathy or hypoalbuminemia, insufficient space for the transplant kidney, primary focal segmental glomerulosclerosis, child with urine volume greater than 2.5 mg/kg/hour, chronic kidney infection in a nonfunctional kidney, or if the disease affecting the native kidney predisposes to increased cancer risk.(1)(3)(14)(20) [C in Context Link 1, 2]

[D] See Clinical Indications for Procedure in this guideline. [D in Context Link 1]

[E] Patient is ambulatory or near baseline activity for age and development. [E in Context Link 1, 2]

[F] Use Multimodal analgesia or individual analgesic agent as indicated. [F in Context Link 1]

[G] Patient will be discharged on immunosuppressive medication and corticosteroids. [G in Context Link 1]

[H] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [H in Context Link 1]

[I] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(38) [I in Context Link 1]

Definitions

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
  - Hemodynamic abnormalities at baseline or acceptable for next level of care
  - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
    - Tachycardia absent
    - Hypotension absent
    - No evidence of inadequate perfusion (eg, no myocardial ischemia)
    - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

References

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Hypotension absent

- Hypotension absent[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
  - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
  - Mean arterial pressure[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
  - Mean arterial pressure[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
  - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
  - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

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Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP.

Multimodal analgesia

- Multimodal analgesia involves the utilization of 2 or more analgesic agents with different mechanisms of action in order to provide additive or synergistic pain control, while minimizing side effects and reliance on opioids.(1)(2)(3)

References

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2. George S, Johns M. Review of nonopioid multimodal analgesia for surgical and trauma patients. American Journal of Health-System Pharmacy 2020;77(24):2052-2063. DOI: 10.1093/ajhp/zxaa301.
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Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
  - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
  - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

References

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2. Dalal AS, Van Hare GF. Syncope. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:566-571.e1.
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Footnotes

A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)

B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):
  - Housing insecurity, as indicated by **1 or more** of the following:
    - Individual or caregiver's current living situation is **1 or more** of the following(5):
      - Does not have own housing (eg, staying in a hotel, shelter, or with others)
      - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
      - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
    - Current housing has **1 or more** of the following:
      - Electrical appliances (eg, stove, refrigerator) not working or unavailable
      - Insufficient heating or cooling
      - Insufficient ventilation
      - Lead paint or pipes
      - Mold
      - Pests (eg, bugs) or rodents
      - Smoke detectors not working or unavailable
  - Food insecurity, as indicated by **1 or more** of the following(6):
    - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
    - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
  - Insufficient transportation, as indicated by **1 or more** of the following(7):
    - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
    - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.

- Insufficient utilities, as indicated by **1 or more** of the following(8):
  - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
  - In the past year, electric, water, gas, or oil company threatened to shut off services.
- Personal safety risk, as indicated by **2 or more** of the following(6):
  - Individual is sometimes or frequently physically hurt by another person (including family member).
  - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
  - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
  - Individual is sometimes or frequently screamed or cursed at by another person (including family member).
- Insufficient dependent care, as indicated by **1 or more** of the following:
  - In the past year, individual or caregiver was unable to work due to lack of dependent care.
  - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
  - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
  - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
- Depression risk, as indicated by **ALL** of the following:
  - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
  - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

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1. Scanlon A, Reinisch C. Social determinants of health. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:18-20.
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Tachycardia absent

- Tachycardia[A][B] absent, as indicated by **1 or more** of the following(1)(2):
  - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
  - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
  - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
  - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
  - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
  - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

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Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Codes

ICD-10 Diagnosis: N18.5, N18.6, Q61.19, Q61.2, Q61.3, Q61.5, T86.10, T86.11, T86.12, T86.13, T86.19

ICD-10 Procedure: 0TY00Z0, 0TY00Z1, 0TY00Z2, 0TY10Z0, 0TY10Z1, 0TY10Z2

CPT®: 50360, 50365

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